

RFP Response

VMware to KVM Migration Services

Prepared for: [Customer Organization Name]

Prepared by: [Your Company Name]

Date: [Date]

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[Insert Your Company Logo Here]

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1. Executive Summary

[Your Company] is pleased to respond to [Customer]'s Request for Proposal for VMware to KVM migration services. We propose a comprehensive migration of [N] virtual machines from VMware vSphere to KVM-based infrastructure using our proven methodology and the open-source hyper2kvm migration toolkit.

Key Value Proposition:

- Eliminate VMware/Broadcom licensing costs: estimated \$XXX,XXX /year savings
- Zero-cost migration tooling (hyper2kvm — Apache-2.0 open source)
- Proven methodology with automated conversion, testing, and rollback
- Target completion: [X] weeks with less than 4 hours downtime per VM
- Post-migration support and knowledge transfer included

2. Company Qualifications

Q: Describe your company's experience with virtualization migration projects.

[Your Company] has completed [N] virtualization migration projects over the past [N] years, migrating a total of [N] virtual machines across industries including [industries]. Our team holds certifications including [RHCSA, CKA, VCP, etc.] .

Q: What migration tools do you use?

We use **hyper2kvm** (h2kvmctl), an open-source migration toolkit licensed under Apache-2.0. Key capabilities:

- **10 source formats:** VMware VMDK, OVA, OVF, Hyper-V VHD/VHDX, AWS AMI, Azure VHD, GCP images, and raw disk images
- **Automated fixes:** VirtIO driver injection (Windows), initramfs rebuild (Linux), fstab UUID conversion, GRUB repair
- **Dual deployment:** libvirt/KVM and KubeVirt (Kubernetes) in a single workflow
- **YAML-driven:** Declarative batch migration configs with dependency ordering
- **Zero cost:** No per-VM fees, no license keys, no vendor lock-in

Q: What certifications does your team hold?

Certification		Team Members	Relevance
RHCSA / RHCE	[N]	Linux system administration	
CKA / CKAD	[N]	Kubernetes administration	
VCP-DCV	[N]	VMware source environment expertise	
AWS SAA / Azure AZ-104	[N]	Cloud-to-KVM repatriation	

3. Technical Approach

Q: Describe your technical approach to converting VMware VMs to KVM format.

Our conversion pipeline uses hyper2kvm's VMCraft engine (pure Python + qemu-nbd), which processes each VM through these stages:

- **Export:** govc-based vSphere export with streaming progress monitoring
- **Convert:** VMDK/OVA → QCOW2 with optional compression (40-60% space savings)
- **Fix:** Automated offline modifications — VirtIO drivers (Windows), initramfs rebuild (Linux), fstab UUID conversion, GRUB update, network configuration
- **Deploy:** Generate libvirt domain XML with NUMA-aware CPU pinning, hugepages, VirtIO devices, and correct UEFI/BIOS firmware detection
- **Validate:** Automated boot test, service verification, and performance baseline comparison

No manual intervention required for standard Linux and Windows VMs. Custom handling documented for edge cases (vendor appliances, legacy OS).

Q: How do you handle Windows VMs and prevent BSOD?

Windows BSOD (0x7B INACCESSIBLE_BOOT_DEVICE) is the #1 migration failure mode. h2kvmctl prevents this by:

- Mounting the guest disk offline using qemu-nbd
- Loading the Windows SYSTEM registry hive
- Injecting VirtIO storage drivers (viosstor/vioscsi) as boot-critical
- Injecting VirtIO network drivers (NetKVM) and balloon drivers
- Supporting Windows Server 2012 R2 through 2025, and Windows 10/11

We carry the latest **virtio-win.iso** (from Fedora) with pre-tested drivers for all supported Windows versions. **Zero BSOD guarantee** with our process.

Q: What target KVM platforms do you support?

Platform	Use Case	Scale
KVM + libvirt (RHEL/Rocky/Ubuntu)	Traditional VM hosting, direct VMware replacement	Any
KubeVirt on K3s	Edge, small deployments, container+VM convergence	1-50 VMs
KubeVirt on RKE2/Vanilla K8s	Enterprise Kubernetes with VM support	50-500 VMs
OpenShift Virtualization	Enterprise with Red Hat support	100+ VMs
oVirt / OLVM	VMware-like management experience	50-500 VMs

4. Migration Methodology

Q: Describe your migration methodology and phases.

Our proven 5-phase methodology:

Phase	Duration	Activities	Deliverables
1. Discover	1-2 weeks	VM inventory, dependency mapping, risk assessment, workload classification	Migration plan, VM matrix, risk register
2. Design	1 week	Target architecture, network design, storage sizing, HA/DR planning	Architecture document, sizing calculator output
3. Build	1-2 weeks	Deploy KVM hosts, configure networking/storage, install monitoring	Production-ready KVM infrastructure
4. Migrate	2-6 weeks	Wave-based migration: convert, test, cutover, validate per wave	Migrated VMs, test results, sign-off per wave
5. Optimize	2 weeks	Performance tuning, documentation, knowledge transfer, VMware decommission	Runbooks, training, PIR report

Q: How do you handle rollback if a migration fails?

Rollback is immediate and zero-risk:

- Source VMware VMs are **never modified or deleted** during migration
- VMware VMs remain powered off (not deleted) for 48 hours after cutover
- Rollback = power on VMware VM, power off KVM VM (2-minute process)
- MAC addresses are preserved, so network/DHCP/DNS require no changes
- Decision deadline: 2 hours after go-live per wave
- We have achieved **0% rollback rate** across our last [N] migration projects due to thorough pre-testing

Q: What is the expected downtime per VM during migration?

Typical downtime: 5-15 minutes per VM. Our process pre-converts VMs while they are still running on VMware. The cutover window involves only: (1) clean shutdown of VMware VM, (2) boot KVM VM. The conversion itself runs hours or days before the cutover, so it does not contribute to downtime. For large databases requiring consistency, we recommend a 2-4 hour window that includes final backup verification.

5. Tooling & Automation

Q: What licensing costs are associated with your migration tools?

\$0. hyper2kvm (h2kvmctl) is licensed under Apache-2.0 — free for commercial use with no per-VM fees, no license keys, and no usage limits. The customer retains the tool post-migration for future use. All underlying components (QEMU, libvirt, KVM) are open-source and included in standard Linux distributions.

Q: Can the migration be automated for large-scale environments?

Yes. h2kvmctl supports:

- **YAML batch configuration:** Define all VMs in a single file with dependency ordering
- **CI/CD integration:** GitLab CI, Jenkins, and GitHub Actions pipeline templates included
- **Ansible playbooks:** Inventory-driven migration across multiple hosts
- **Parallel execution:** Convert 10+ VMs simultaneously (limited by I/O bandwidth)
- **JSON output:** Machine-readable results for custom dashboards and reporting
- **Dry-run mode:** Validate everything before touching production

For a 500-VM environment, automated batch processing reduces per-VM effort from 2 hours (manual) to 6 minutes (automated).

6. Security & Compliance

Q: How do you ensure data integrity during migration?

- **SHA-256 checksums:** Source and destination images are checksummed and compared
- **Audit logging:** Every operation logged with timestamp, operator, action, and result (JSON structured)
- **No data traversal:** Conversion happens locally on the target host — VM data never leaves the data center
- **Encryption:** Optional AES-256-GCM encryption for media transfer in air-gapped environments
- **Evidence packages:** Automated generation of compliance-ready evidence for SOX, HIPAA, PCI-DSS

Q: What compliance frameworks have you supported in previous migrations?

Framework	Support Level	Key Controls
SOX (ITGC)	Full	Change management, separation of duties, audit trail
HIPAA	Full	ePHI protection, access controls, encryption, BAA-compatible
PCI-DSS v4.0	Full	Change control (6.4), audit trail (10.2), vulnerability scanning (11.2)
SOC 2 Type II	Full	Change management (CC8.1), monitoring, logical access
FedRAMP / NIST 800-53	Full	Configuration management (CM-3), access control (AC-2)
FIPS 140-2	Full	FIPS-validated crypto for air-gapped environments

7. Support & SLA

Q: What post-migration support do you provide?

Support Tier	Duration	Coverage	Response Time
Hypercare	First 2 weeks post-migration	24/7 on-call, dedicated engineer	15 min (critical), 1 hr (high)
Standard	Months 1-3	Business hours, remote support	1 hr (critical), 4 hr (high)
Extended (optional)	Months 4-12	Business hours, advisory	4 hr (critical), 1 day (standard)

Q: What knowledge transfer is included?

- **Documentation:** Complete runbooks for KVM administration (virsh commands, networking, storage, backup, troubleshooting)
- **Training:** [N] days of hands-on training for your team (VM lifecycle, monitoring, Pacemaker HA)
- **Lab exercises:** 10 guided labs covering all Day-2 operations
- **Video recordings:** All training sessions recorded for future reference
- **Office hours:** Weekly 1-hour Q&A sessions during first month

8. Timeline & Milestones

Q: Provide a proposed project timeline.

Milestone	Target Date	Deliverable	Gate
Project kickoff	Week 1	Kick-off meeting, access granted	SOW signed
Discovery complete	Week 2	VM inventory, migration plan	Plan approved
Infrastructure ready	Week 4	KVM hosts deployed, tested	Doctor checks pass
POC complete (5 VMs)	Week 5	5 VMs migrated, validated	POC sign-off
Wave 1 (dev/test)	Week 7	Dev/test VMs migrated	All boot tests pass
Wave 2 (databases)	Week 9	Database VMs migrated	App owner sign-off
Wave 3 (production)	Week 11	All production VMs migrated	All smoke tests pass
VMware decommission	Week 13	ESXi hosts powered off	48-hour soak complete
Project close	Week 14	PIR, runbooks, training complete	Customer sign-off

Timeline assumes [N]

VMs. Adjust proportionally for larger environments.

9. Pricing

Q: Provide a detailed cost breakdown.

Line Item	Unit	Quantity	Unit Price	Total
Discovery & Assessment	Fixed	1	\$X,XXX	
			\$X,XXX	
Architecture Design	Fixed	1	\$X,XXX	
			\$X,XXX	
Infrastructure Build	Per host	[N]		
		\$X,XXX		
		\$XX,XXX		
VM Migration (conversion + testing)	Per VM	[N]		
		\$XXX		
		\$XX,XXX		
Cutover Support (weekend windows)	Per wave	[N]		
		\$X,XXX		
		\$XX,XXX		
Knowledge Transfer & Training	Fixed	1	\$X,XXX	
			\$X,XXX	
Hypercare Support (2 weeks)	Included	1	Included	\$0
hyper2kvm Software License	Per project	1	\$0	\$0
TOTAL PROJECT COST				\$XXX,XXX

Customer ROI Summary:

Annual VMware licensing eliminated: \$XXX,XXX /yr

Migration project investment: \$XXX,XXX (one-time)

Payback period: [N] months

5-year net savings: \$X.XM

10. References & Case Studies

Customer	Industry	VMs Migrated	Annual Savings	Contact
[Reference 1]	[Industry]			
[N]	\$XXX,XXX			
[Name, Title, Email]				
[Reference 2]	[Industry]			
[N]	\$XXX,XXX			
[Name, Title, Email]				
[Reference 3]	[Industry]			
[N]	\$XXX,XXX			
[Name, Title, Email]				

Appendix A: Detailed Technical Q&A

Q: What disk formats are supported for conversion?

VMDK (all sub-formats: monolithic sparse, monolithic flat, split sparse, stream-optimized), OVA, OVF+VMDK, VHD, VHDX, RAW, QCOW2. Source format is auto-detected.

Q: How are VM network configurations preserved?

MAC addresses are preserved from OVF metadata, ensuring DHCP leases, license servers, and monitoring continue without changes. Network topology (VLAN assignments) is mapped to Linux bridge or OVS configurations. h2kvmctl auto-generates libvirt network XML from OVF NetworkSection.

Q: Can you migrate VMs with snapshots?

Yes. VMware snapshot chains are consolidated during export (govc export produces a flat image). The resulting image contains the latest state including all snapshots. We recommend consolidating snapshots in VMware before export for fastest processing.

Q: How do you handle UEFI vs BIOS boot VMs?

h2kvmctl auto-detects firmware type by examining the partition table (GPT = UEFI, MBR = BIOS) and OVF metadata. UEFI VMs use OVMF firmware (edk2-ovmf package). Secure Boot is supported with enrolled keys. Windows 11 VMs additionally get a virtual TPM 2.0 (swtpm).

Q: What monitoring replaces vCenter after migration?

We deploy Prometheus + Grafana + Alertmanager (all open-source, \$0 cost) with pre-built dashboards for KVM host metrics, per-VM resource usage, capacity forecasting, and alerting. Optional: Loki for log aggregation (replaces vRealize Log Insight). Web-based management via Cockpit (included with RHEL/Rocky).

Q: How does KVM high availability compare to vSphere HA?

Pacemaker + Corosync provides equivalent automatic VM restart on host failure. With shared storage (Ceph or NFS), VMs are restarted on surviving hosts within 30-60 seconds. Live migration (virsh migrate --live) provides vMotion-equivalent functionality with sub-100ms downtime. Both 2-node (with quorum device) and 3+ node clusters are supported.

Q: Can you migrate to KubeVirt (Kubernetes) instead of standalone KVM?

Yes. h2kvmctl supports direct deployment to KubeVirt via the CDI (Containerized Data Importer). Converted QCOW2 images are uploaded as PersistentVolumeClaims, and VirtualMachine resources are created in the target namespace. This enables running VMs and containers on the same Kubernetes cluster with unified kubectl management, monitoring, and networking.

Q: What is the ongoing cost of running KVM vs VMware?

KVM (kernel module) and libvirt are included in every Linux distribution at no cost.
Optional: RHEL subscription (\$799/host/year) for vendor support, or use Rocky Linux / AlmaLinux for \$0. There are no per-VM fees, no per-core fees, no per-socket fees, and no forced bundles. Hardware upgrades (more cores, more RAM) never increase software costs.

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